

Design of a Dual Automatic Door Control System with Optocoupler Isolation and Docking Charging Mechanism Based on Arduino Mega and ESP32

This report was prepared to fulfill one of the requirements for completing a
Diploma Four in Telecommunications Engineering.

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APPROVAL SHEET

**Design and Development of a Dual Automatic Door Control
System with Optocoupler Isolation and Docking Charging
Mechanism Based on Arduino Mega and ESP32**

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AUTHOR'S STATEMENT

Hereby states that the Field Work Practice report titled "Design and Construction of a Dual Automatic Door Control System with Optocoupler Isolation and Docking Charging Mechanism Based on Arduino Mega and ESP32" is free from any elements of plagiarism and complies with the applicable writing guidelines.

Should any elements of plagiarism be discovered in the future, the assessment results of this Field Work Practice will be revoked, and I am willing to accept sanctions in accordance with applicable regulations.

This statement is made truthfully and in full awareness.

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FOREWORD

The author would like to express his gratitude to Allah SWT, for it is through His grace and blessings that this internship report entitled Design and Construction of a Dual Automatic Door Control System with Optocoupler Isolation and Arduino Mega and ESP32-based Docking Charging Mechanism could be completed on time.

This report was prepared as a form of accountability for the internship activities that were carried out during a certain period. Through these activities, the author gained a lot of experience, knowledge, and new understanding about Micro Controllers.

In the process of compiling this report, the author fully realizes that the success of the internship is inseparable from the support, guidance, and assistance from various parties. Therefore, the author would like to express his deepest gratitude to:

1. Mr. Ir. Hertog Nugroho, M.Sc., Ph.D. as the academic supervisor who provided direction, guidance, and motivation during the internship.
2. Mr. Fiktor Adil Rahmat Ndruru as the field supervisor who provided opportunities, knowledge, and direct guidance during the internship activities.
3. PT Solusi Intek Indonesia for providing the opportunity and facilities to carry out the internship activities, enabling the author to gain real-world work experience.
4. All staff and employees of PT Solusi Intek Indonesia who have assisted and provided valuable experiences throughout the internship period.
5. Parents and family who have always provided prayers, support, and encouragement, enabling the author to complete this activity and report successfully.

The author acknowledges that this report is far from perfect, both in terms of content and presentation. Therefore, constructive criticism and suggestions are greatly

appreciated for the improvement of future reports. May this report be of benefit, both to the author and to those who may need it.

Bandung, ...

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BAB I

INTRODUCTION

I.1 Background

The development of automation technology has provided convenience in various fields, including mechanical control systems and mobile devices such as drones. In the context of modern industry, the need for automatic door control systems that are safe, precise, and can be integrated with other technologies is increasing. This is especially important in applications such as drone docking stations, where doors must open and close in a coordinated manner to protect the device and allow for safe entry, exit, and charging.

To meet these requirements, a control system is needed that can accurately regulate motor movement, monitor boundary conditions through limit switch sensors, and provide an easy-to-use close-range control and monitoring mechanism. Integrating microcontrollers such as Arduino Mega for motor control and ESP32 as a web-based interface provides an efficient and flexible solution. Through this approach, the automation process can be carried out in a more structured, measurable, and adaptable manner for various light industrial applications and automation technology development prototypes.

I.2 Objectives

The objective of this internship project is to design and implement a dual automatic door system integrated with a drone docking mechanism. Specifically, the objectives of this system design are as follows:

1. Designing and building an Arduino Mega (ATmega2560)-based automatic double door control system capable of controlling the movement of two stepper motors synchronously or separately according to commands, with precise time delay settings.
2. Implementing a limit switch as a position protection system, so that the motor can stop automatically when the door reaches the open or closed limit to prevent mechanical damage and improve operational safety.

3. Applying signal isolation using a PC817 optocoupler to protect the microcontroller from electrical interference, current surges, or noise that may originate from the TB6600 motor driver.
4. Developing an automatic docking charging mechanism that only activates when both doors are completely closed, so that the drone charging process can take place safely, stably, and without interfering with door operations.
5. Implementing non-blocking control logic based on the millis() function, so that all processes such as motor movement, limit switch monitoring, indicator light system, and relay settings can run in parallel without hindering other functions.
6. Integrating ESP32 as an IoT module for close-range control and monitoring, using a web server that displays the system status in real-time and allows users to give commands to open and close doors as well as set up docking charging.
7. Designing an automatic indicator light system that provides a visual response in the form of a blinking light when the door is about to open and a red strip light when the door is about to close, thereby maximizing user safety and comfort.
8. Developing a prototype of a dual automatic door system that can be used as part of a hangar or drone docking system, where the door opens when the drone is about to land or take off, and closes again to protect and automatically charge the drone.

I.3 Benefits

The benefits of the Field Work Practice activity with the topic “Design and Development of a Dual Automatic Door System for Drone Docking Based on Arduino Mega and ESP32” are as follows:

I.3.1 For the Author

- 1) Enhancing technical understanding in microcontroller-based door automation systems, particularly in controlling two stepper motors using Arduino Mega with a non-blocking control method based on millis().

- 2) Gaining practical experience in designing integrated electronic circuits, including the use of PC817 optocouplers as signal isolators and the integration of TB6600/DM542 motor drivers.
- 3) Developing the ability to build a simple IoT system using ESP32, specifically the implementation of a web server for real-time door monitoring and control.
- 4) Enhancing analysis and troubleshooting skills, both in programming and hardware, when handling the synchronization of two doors, limit switch handling, and command processing via serial communication.
- 5) Expanding knowledge on drone docking automation, including how automatic doors and charging systems can work together in a safe and efficient system.

I.3.2 For Companies/Institutions

- 1) Providing a ready-to-use or developable prototype of an automatic door control system, without affecting the mechanical design, so it can be integrated with any door structure.
- 2) Adding value to the development of microcontroller-based automation systems and IoT, which can be applied to drone storage systems, smart gates, or other industrial devices.
- 3) Supporting research and development activities, particularly in the creation of safe, modular control systems that are easy to integrate with any mechanical device.

I.3.3 For Academics/Readers

- 1) Serves as a reference for the design of a dual automatic door control system based on Arduino Mega and ESP32, without focusing on the mechanical design of the door.
- 2) Providing technical insights into stepper motor control, optocoupler isolation, serial communication, and IoT integration, making it a reference for similar research.

- 3) Demonstrates the application of control system and electronics concepts, enabling readers to understand how to design a complete controller that can work with various mechanical door systems.

I.4 Time and Place of Implementation

This internship was conducted on:

Date : July 1 to January 2, 2025

Time : Monday to Saturday

Location : PT Solusi Intek Indonesia, Product Engineer Unit

I.5 Writing System

This Field Work Practice Report is systematically organized so that the discussion can be conveyed clearly and easily understood. **Chapter I Introduction** contains a description of the background of the report preparation, problem formulation, objectives, benefits, and writing system. This chapter provides an initial overview of the reasons for conducting the practice, the focus of the study to be discussed, and the overall direction of the report. **Chapter II PT Solusi Intek Indonesia** contains an explanation of the profile of the institution where the practice was carried out, including a brief history, vision and mission, organizational structure, business fields, and the scope of activities related to the company's work environment. This explanation aims to provide readers with an understanding of the company's operational conditions and their relevance to the practice activities.

Next, **Chapter III Implementation of Field Work Practice** describes the implementation process in more detail. This chapter begins with an overview that explains the scope of activities and the main topics of discussion. The next section presents the relevant theoretical basis for understanding the concepts and technologies used. After that, this chapter

explains the hardware and software that support the implementation of activities, then continues with the steps for implementing the practice, starting from preparation, assembly, testing, to recording the results. In the final section, an analysis of the results obtained is also presented, along with any technical or non-technical obstacles encountered during the implementation of activities.

Chapter IV Conclusion is the final section containing the conclusions of the activities that have been carried out, including achievements, analysis results, and the effectiveness of the processes carried out. In addition, this chapter contains suggestions intended as input for the development of practical activities and improvements to the system that will be the focus of discussion in the future. With a structured layout in these four chapters, this report is expected to provide a clear and comprehensive overview of the Field Work Practice activities.

BAB II

PT SOLUSI INTEK INDONESIA

II.1 Company Description

PT. Solusi Intek Indonesia is a company engaged in the development and application of information technology, particularly Intelligence Information Technology & Intelligence Equipment. Since 2017, PT. Solusi Intek Indonesia has conducted various research and technological developments in collaboration with various parties to produce various Intelligence Information Technology Systems and Equipment that are suitable for conditions in Indonesia.

Indonesia possesses social, political, and cultural diversity, along with numerous border regions. Therefore, PT. Solusi Intek Indonesia aims to support and provide solutions for the execution of intelligence operations in maintaining stability and security in Indonesia.

II.2 Logo of PT. Solusi Intek Indonesia

The following is the official logo of PT. Solusi Intek Indonesia:



Figure II.2.1 PT. Solusi Intek Indonesia Logo

II.3 Company Vision and Mission

Vision:

PT Solusi Intek Indonesia is an Indonesia-based company that aims to be the best partner company in providing intelligence operations that support the implementation of real-time security information in order to maintain the stability and security of the Republic of Indonesia.

Mission:

1. To provide products that deliver the best solutions in the field of Information Technology and Intelligence Equipment.
2. Conducting continuous research and development to produce the best products created by the nation's children.

3. Continuously following the development of intelligence technology in various countries.

II.4 Company Work Principles

PT. Solusi Intek Indonesia is a company engaged in the field of information technology, with a special focus on the development and application of intelligence information technology and intelligence equipment. Since its establishment in 2017, the company has conducted various research and development of technologies designed to meet the operational needs of intelligence in Indonesia, supporting national stability and security through technological innovation.

II.5 Company Structure

PT. Solusi Intek Indonesia has the following company structure:

1. CEO (Chief Executive Officer)
2. COO (Chief Operating Officer)
3. General Manager
4. Commissioner
5. Manager
6. Project Manager
7. Supervisor
8. Staff

PT.Solusi Intek Indonesia also has the following departments/divisions:

1. Human Resources (HR)
2. L A W (Legal)
3. General Affairs (GA)
4. Corporate Social Responsibility (CSR)
5. Health, Safety, and Environment (HSE)
6. Finance Business Development
7. Support
8. Inventory
9. Software Engineering
10. Mechatronics
11. Manufacturing
12. Microcontroller

13. Product Design

14. Product Engineering

II.6 Company Products

1. Anti-Surveillance

Equipment used to detect the installation of eavesdropping devices in a specific room/area. This equipment makes a room or area free from various types of eavesdropping devices.

2. Surveillance

3. A monitoring device used to eavesdrop on an object, behavior, or activity to obtain information. Eavesdropping can be obtained in the form of sound/voice, images, or video obtained inside or outside a room.

4. Tactical Equipment

- IMSI Catcher
- Peer-to-Peer Blaster
- SMS Blaster
- Direction Finder
- Electromagnetic Pulse (EMP)

5. Various Electronic Intelligence (VELINT)

- Geospatial Intelligence Analysis (GIA)
- Intelligence Portal Data (IPD)
- Flight Monitoring System (CAMAR)

6. Open Source Intelligence (OSINT)

- Intelligence Media Analytics (IMA)
- Intelligence Socio Analytics (ISA)
- Intelligence Cybertroops System (ICT)
- Geostrategic Intelligence (GINT)
- Intelligence Admiralty System (IAS)
- Massive Profiling (MASSPRO)

7. Signal Intelligence (SIGINT)

- WhatsApp Group Monitoring System (WGMS)
- Intelligence Profiling System (IPS)
- People Movement Tracking & Analysis (PMTA)

8. Cyber Security
 - Gate Scanner
 - Cyber Range

II.7 Company Location

This field work practice was carried out at Workshop 08 Cikunir, located at Cikunir Raya No.8, RT.004/RW.011, Jatimekar, Kec. Jatiasih, Kota Bks, West Java 17422. The following is the location of the field work practice taken using Google Earth.

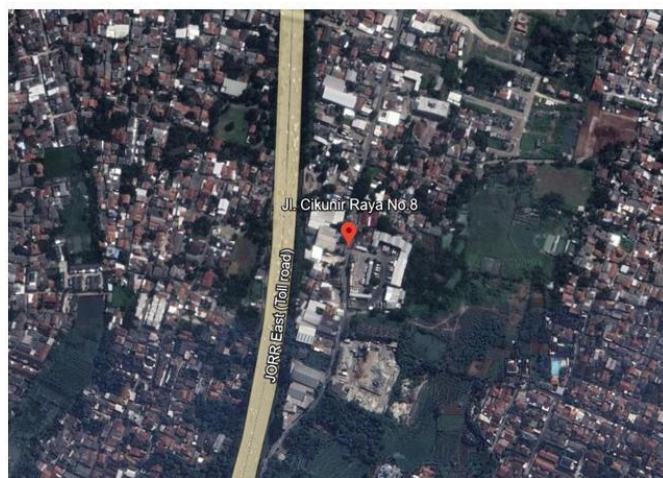


Figure II.7.1 Location of Cikunir Workshop 08



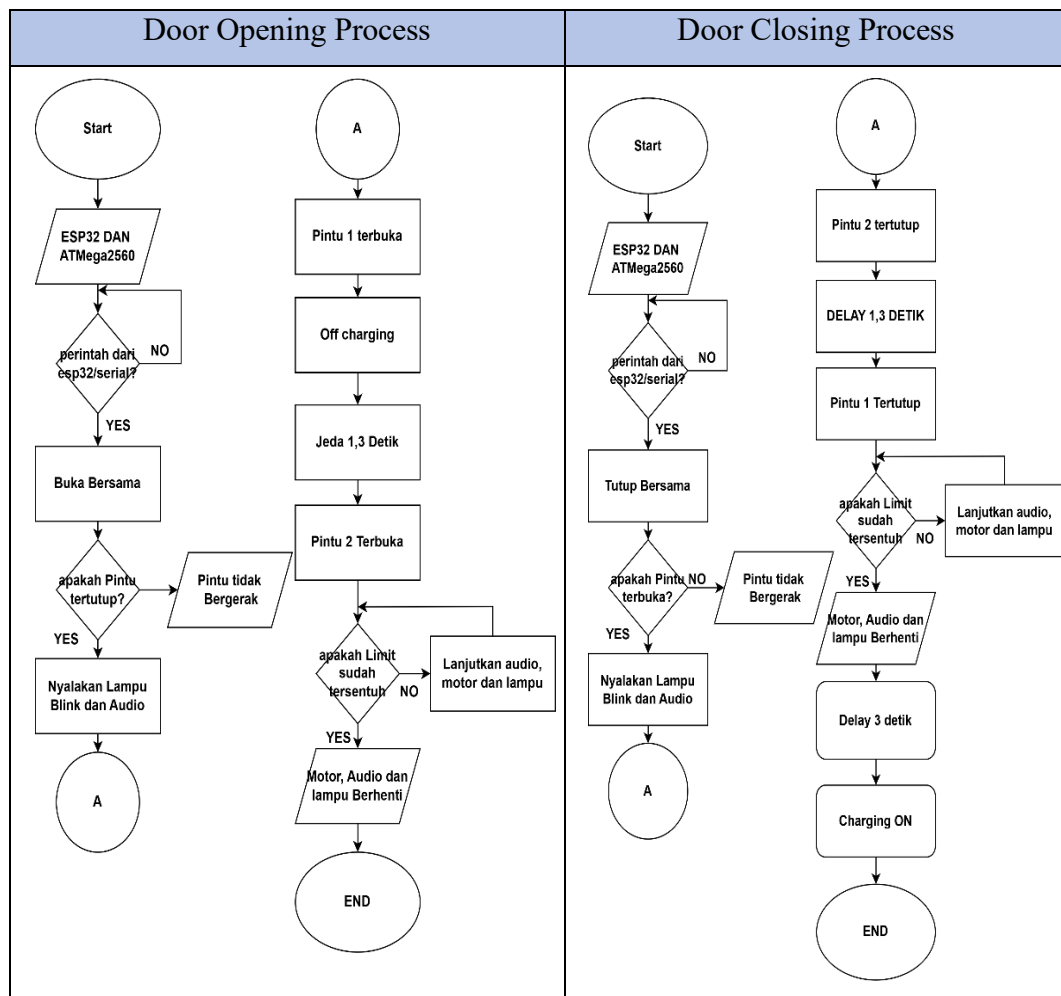
Figure II.7.2Cikunir 08 Workshop

BAB III

DESIGN AND SYSTEM IMPLEMENTATION

III.1 General Overview

This Field Work Practice activity focuses on the design and construction of an electronic control system for double automatic doors at a drone docking station. The main objective of this design is to provide an automatic door controller that is capable of working precisely, synchronously, and safely without designing the mechanical aspects of the door. This system is intended for hangar or drone docking needs, where the doors must be able to open and close automatically according to the drone's operating conditions. With proper control, the doors can provide access when the drone is landing and taking off, and close safely again when the drone is stored in the docking station.

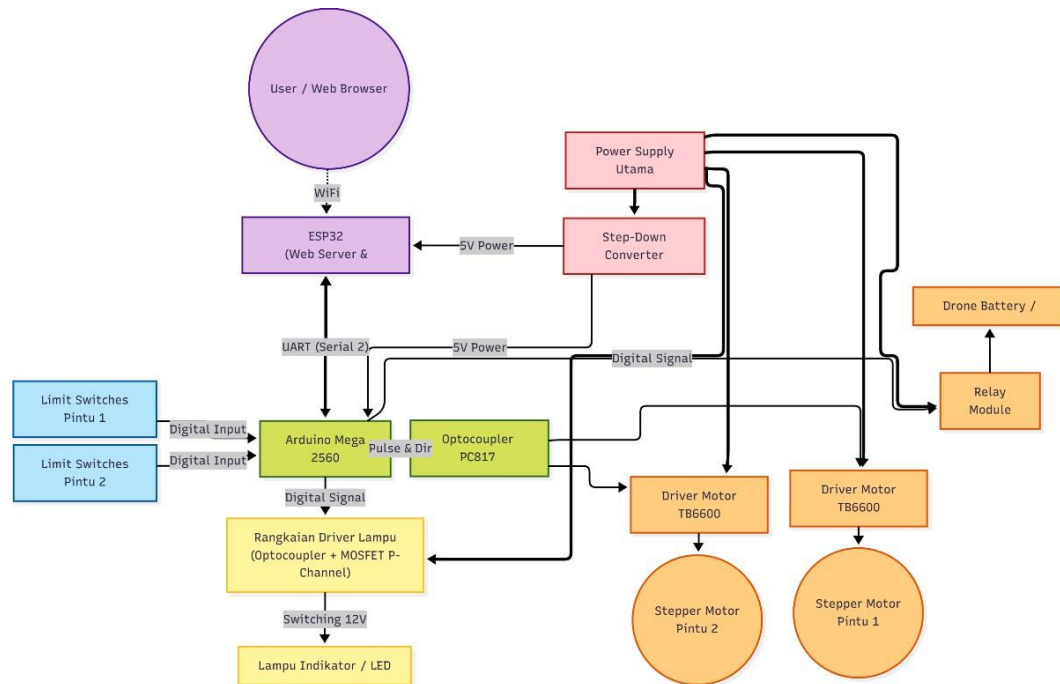


This control system was developed using an Arduino Mega (ATmega2560) microcontroller as the main control center and ESP32 as a web server-based interface module that enables wireless monitoring and operation of the door. The entire implementation process was carried out in a laboratory and electronics workshop environment that supports the design, assembly, and testing of microcontroller-based control circuits.

The door control system uses two stepper motors driven by a TB6600 driver, with open-close logic implemented based on a non-blocking control method using the `millis()` function. This technique allows both motors, limit switch sensors, optocouplers, and relay modules to run in parallel without any obstacles due to the use of `delay()`, making the system response faster and more stable.

To ensure safety and prevent damage to the door mechanism, limit switches are used as open and close limit sensors. These sensors act as interlocks, ensuring that the motors stop automatically when the doors reach their end positions. In addition, PC817 optocouplers are used as optical isolation to protect the Arduino Mega from electrical interference, current surges, and noise generated by the motor drivers.

One of the main features of this control system is automatic docking charging, which is a charging relay activation mechanism that only works when both doors are fully closed. This ensures that the drone charging process is safe and free from the risk of mechanical or electrical interference. On the interface side, ESP32 is used to provide a control web server, so that users can send open, close, stop, and activate or deactivate charging commands via a Wi-Fi network, while monitoring the status of the doors and sensors in real-time.

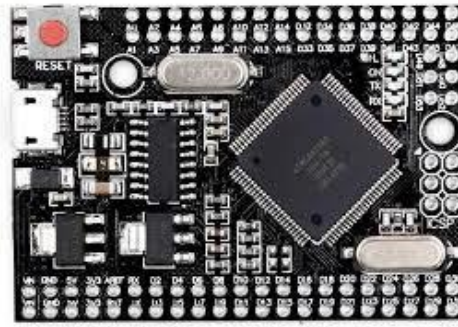


All of these activities provide practical experience in designing an integrated electronic control circuit, from the circuit design stage, program writing, motor and sensor driver testing, to the creation of a web-based interface system. This chapter will explain in detail the hardware and software used, the design flow, and the testing methods to ensure that the dual automatic door control system can work optimally and reliably.

III.2 Related Theoretical Foundations

III.2.1 Arduino Mega Microcontroller (ATMega2560)

The Arduino Mega 2560 is an ATMega2560-based microcontroller board that is widely used in control applications because it has more input/output pins than other Arduino types. This microcontroller provides 54 digital pins, 16 analog pins, and 4 UART ports, making it ideal for systems that require multidevice communication and control of a large number of actuators [1].



In this project, Arduino Mega acts as the main control unit that regulates direction signals (DIR), step pulses (STEP/PUL), limit switch readings, and docking charging relay operations. The control logic implementation uses a non-blocking method based on the `millis()` function so that all processes run in parallel without hindering other tasks.

III.2.2 Stepper Motor

A stepper motor is an electric motor that moves based on the number of steps per electrical pulse, enabling it to produce precise positions and speeds. Stepper motors typically have a step angle of 1.8° per step or 200 steps per full rotation [2].



This system uses a HUIYINN 57DNG508Y-21D20 stepper motor, which has high torque and is suitable for automatic door applications. The motor is driven by PUL (pulse) and DIR (direction) signals from the microcontroller, allowing accurate control of door movement during both opening and closing.

III.2.3 TB6600 Motor Driver

The TB6600 driver functions as a current amplifier and bridge between the Arduino logic signal and the stepper motor [3]. Arduino is unable to provide direct current to drive the motor due to its limited output current, so an external driver is required.



The TB6600 operates based on STEP (pulses) and DIR (direction) signals. Additionally, this driver supports microstepping mode, which makes motor movement smoother. In this project, microstep mode is used to produce stable door movement with minimal vibration.

III.2.4 Limit Switch

A limit switch is a mechanical sensor that detects the linear or rotational movement limits of a mechanism [4]. Its working principle is a change in contact condition when the switch lever is pressed by a mechanical object. There are two main types, namely Normally Open (NO) and Normally Closed (NC).



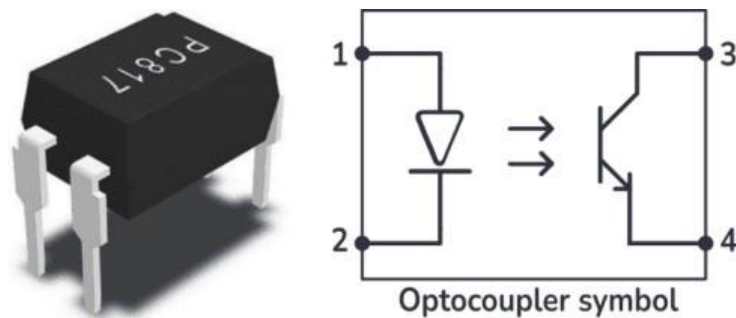
In the automatic drone docking door system, two limit switches are installed on each door:

- Forward Limit Switch → detects the fully open position,
- Reverse Limit Switch → detects the fully closed position.

The limit switches are connected to the Arduino using a pull-up input configuration, so that when the button is pressed, the signal changes from HIGH to LOW. This signal is used to automatically stop the motor when the limit position is reached.

III.2.5 OptoCoupler (PC817)

An optocoupler or optical isolator is an electronic component used to separate two electrical circuits using light [5]. This component contains an infrared LED and a phototransistor in a single package. When the LED is turned on, the light activates the phototransistor, allowing the signal to be transmitted without a direct electrical connection.

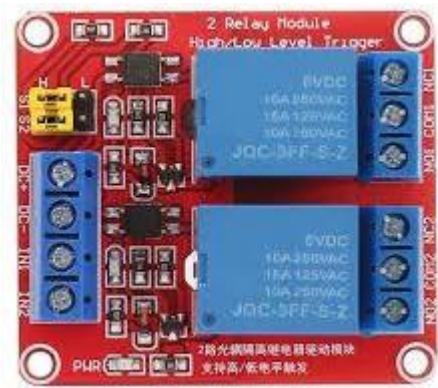


In this project, the **PC817** is used to isolate the logic signal from the Arduino Mega (5 V) from the high voltage on the TB6600 driver side (up to 30 V). Thus, the optocoupler functions to:

- Protecting the microcontroller from back-EMF surges,
- Reducing electrical noise on the signal path,
- Ensuring the long-term safety of the control system.

III.2.6 Relay Docking Charging

A relay is an electromagnetic switch that can be controlled using a small electrical signal to regulate a large current [6]. Relays work based on the principle of electromagnetic induction: when the coil is energized, the resulting magnetic field attracts the mechanical contacts to connect or disconnect the current on the output side.



In this system, two relays are used to connect the positive and negative power supply lines to the drone battery charging terminals. Arduino controls these two relays based on the limit switch status:

- The relay is active when both doors are fully closed (automatic mode),
- The relay can also be controlled manually via serial commands (CHARGE_ON, CHARGE_OFF, CHARGE_AUTO).

This docking relay enables automatic drone charging after the doors are closed, making the system efficient and safe against short circuits.

III.2.7 Non-Blocking Control Principle

In microcontroller-based systems, the use of the `delay()` function often causes other processes to pause temporarily because the CPU waits for the time to elapse. To overcome this, a non-blocking control method is used with the `millis()` function, which calculates time based on the internal timer system [7].

This method enables the program to perform pseudo-multitasking, such as:

- Moving two stepper motors simultaneously,
- Reading the status of a limit switch in real-time,
- Controlling the interval between doors (e.g., 2 seconds),
- Setting parallel charging relays.

With non-blocking logic, the system becomes more responsive, efficient, and synchronized in controlling all devices simultaneously.

III.2.8 ESP32 as a Web Server & Monitoring



ESP32 is a microcontroller module with integrated Wi-Fi and Bluetooth features [8] In this system, ESP32 does not function as a motor controller, but as a web-based user interface.

Its roles include:

- Creating an internal access point (AP)
- Providing a web page for door control
- Sending commands to the Arduino Mega via UART
- Receiving and displaying the system status in real-time

With ESP32, the system can be controlled wirelessly, facilitating device monitoring and testing.

III.3 Hardware and Software Used

- 1) Arduino Mega 2560
- 2) TB6600 Stepper Driver
- 3) HUIYINN 57DNG508Y-21D20 Stepper Motor
- 4) PC817 Optocoupler
- 5) Limit Switch (NO + COM)
- 6) Relay Module (2 Channel)
- 7) ESP32 (Web Server & UI Monitoring)
- 8) 12V DC Power Supply
- 9) Arduino IDE

III.4 Software used

In the dual automatic door control system with a docking charging mechanism for drones, there are two main software programs used, namely Arduino IDE and ESP32-based Web Dashboard. These two software programs play an important role in the design, testing, and overall operation of the system.

Arduino IDE is used as the main platform for writing, compiling, and uploading programs to the Arduino Mega 2560 and ESP32 microcontrollers. The programming language used is C/C++, which is integrated with various supporting libraries such as `HardwareSerial`, `WiFi.h`, and `AsyncWebServer`. The program on the Arduino Mega is responsible for managing the core logic of the system, including stepper motor control, limit switch reading, docking charging relay settings, and indicator light control. The system is designed using a non-blocking timing method based on the `millis()` function, so that each process can run in parallel without hindering other processes. In this way, the system response becomes faster, more stable, and capable of handling multiple tasks simultaneously.



```
aduio213 - command_handler.cpp | Arduino 1.8.19 (Windows Store 1.8.57.0)
File Edit Sketch Tools Help

aduio213 command_handler.cpp $ command_handler.h kontrol_audio.cpp kontrol_audio.h

#include "command_handler.h"
#include "motor_control.h"
#include "limit_switch.h"
#include "relay_control.h"
#include "kontrol_lampu.h"
#include "kontrol_audio.h"

bool bukaBersamaAktif = false;
bool tutupBersamaAktif = false;
bool motor2SudahMulai = false;
unsigned long startDelayTime = 0;
const unsigned long jedaPintu = 1300;

unsigned long limit1TrigTime = 0, limit2TrigTime = 0;
bool limit1DelayActive = false, limit2DelayActive = false;
const unsigned long limitDelay = 70;

bool lampuAudioDelayBukaAktif = false;
bool lampuAudioDelayTutupAktif = false;
unsigned long lampuAudioDelayStart = 0;
const unsigned long lampuAudioDelayDurasi = 2000;

bool delayBukaBersamaAktif = false;
bool delayTutupBersamaAktif = false;
unsigned long delayBersamaStart = 0;
const unsigned long delayBersamaDurasi = 5000;

bool audioSedangBerjalan = false;

bool testSequenceAktif = false;
int testSequenceCounter = 0;
int testSequenceTarget = 100;
enum TestState { TEST_IDLE, TEST_BUKA, TEST_WAIT_BUKA, TEST_TUTUP, TEST_WAIT_TUTUP };
TestState testState = TEST_IDLE;
unsigned long testWaitStart = 0;
const unsigned long testWaitDurasi = 3000;

void handleSerialCommand() {
```

Meanwhile, the ESP32 is used to build a web-based control interface (web dashboard) that can be accessed via a Wi-Fi network. The software on the ESP32 runs an HTTP server using the ESPAsyncWebServer library. Through this dashboard, users can send commands such as OPEN_TOGETHER, CLOSE_TOGETHER, OPEN_1/2, STOP, and CHARGE_ON/OFF. In addition to control functions, the dashboard also displays real-time system status in the form of motor conditions, limit switches, and docking charging conditions. This data is sent from the Arduino Mega to the ESP32 via UART communication and displayed in a browser using JavaScript.

```
coba_esp_bagus213 | Arduino 1.8.19 (Windows Store 1.8.57.0)
File Edit Sketch Tools Help

coba_esp_bagus213

</head><body>
<h2><img alt="Panel icon" data-bbox="348 178 362 188"/> Panel Kontrol & Monitor Pintu Otomatis</h2>

<div>

<button onclick="kirim('BUKA_BERSAMA') ">BUKA BERSAMA</button>
<button onclick="kirim('TUTUP_BERSAMA') ">TUTUP BERSAMA</button>
<button onclick="kirim('STOP') " class="btn-stop">STOP</button><br>
<button onclick="kirim('BUKA_1') ">BUKA 1</button>
<button onclick="kirim('TUTUP_1') ">TUTUP 1</button>
<button onclick="kirim('BUKA_2') ">BUKA 2</button>
<button onclick="kirim('TUTUP_2') ">TUTUP 2</button><br>
<button onclick="kirim('CHARGE_ON') ">CHARGE ON</button>
<button onclick="kirim('CHARGE_OFF') ">CHARGE OFF</button>
<button onclick="kirim('CHARGE_AUTO') ">CHARGE AUTO</button>

</div>

<div class="test-section">
<h3><img alt="Test icon" data-bbox="348 328 362 338"/> Test Sequence</h3>
<button onclick="kirim('TEST_100') " class="btn-test">Test 100x</button>
<button onclick="kirim('TEST_50') " class="btn-test">Test 50x</button>
<button onclick="kirim('TEST_10') " class="btn-test">Test 10x</button><br><br>
<input type="number" id="testCount" value="100" min="1" max="1000">
<button onclick="testCustom() " class="btn-test">Test Custom</button><br><br>
<button onclick="kirim('TEST_STOP') " class="btn-stop"><img alt="Stop icon" data-bbox="398 380 412 390"/> STOP TEST</button>
</div>

<h3>Status Sistem</h3>
<div class="panel">
<div class="indicator"><div id="ledMotor1" class="led"></div>Motor 1</div>
<div class="indicator"><div id="ledMotor2" class="led"></div>Motor 2</div>
<div class="indicator"><div id="ledFwd1" class="led"></div>Limit FWD1</div>
<div class="indicator"><div id="ledRev1" class="led"></div>Limit REV1</div>
<div class="indicator"><div id="ledFwd2" class="led"></div>Limit FWD2</div>
<div class="indicator"><div id="ledRev2" class="led"></div>Limit REV2</div>
<div class="indicator"><div id="ledCharge" class="led"></div>Charging</div>
</div>

<h3><img alt="Log icon" data-bbox="348 505 362 515"/> Log Mega (Realtime)</h3>
<div id="logBox">Menunggu data...</div>

</body>
```

The integration between Arduino Mega, ESP32, and the web dashboard makes the entire system interactive, flexible, and easy to operate. With the support of this software, the door opening and closing mechanism can run precisely, the docking charging process is safe, and system monitoring can be carried out without the need for additional devices. This software ensures that all components work synchronously, stably, and in accordance with the operational needs of the automatic dock system for drones.

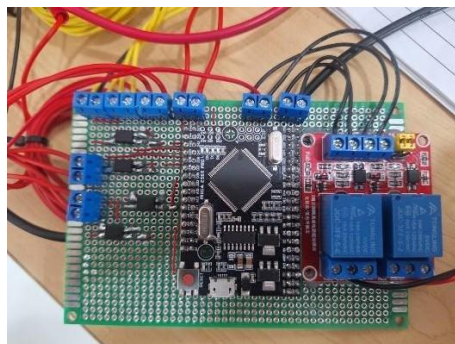
III.5 System Implementation Process

This Field Work Practice activity focused on the design and implementation of an automatic door system with a charging docking mechanism for a drone's

controlled using an Arduino Mega 2560 microcontroller, TB6600 motor driver, and PC817 optocoupler as a signal isolator. The work process was carried out in stages, starting from hardware assembly to overall system testing. Ensure that the room conditions are appropriate:

1) Preparation and Assembly of the Circuit

The first stage begins with preparing all the electronic components used to build the Arduino Mega and ESP32-based dual automatic door control system. These components include the Arduino Mega 2560 as the motor control center, the TB6600 stepper driver to move the two HUIYINN 57DNG508Y-21D20 motors, PC817 optocouplers for signal isolation, relay modules for the charging docking system, limit switches as mechanical limit sensors, and a 12–30 V power supply for the motors and actuator circuit. Once the components are assembled, they are mounted on the PCB board, starting with soldering the headers, PC817, and terminal blocks to make the control circuit more robust and secure. Cable installation continued by connecting the PUL and DIR pins from the Arduino Mega to the two TB6600 drivers via the optocoupler isolation line, as well as connecting the limit switch and relay to the Arduino input pins. At this stage, the ESP32 was also installed separately as a web server module that would later send serial commands to the Arduino Mega via UART communication on the TX2/RX2 pins.

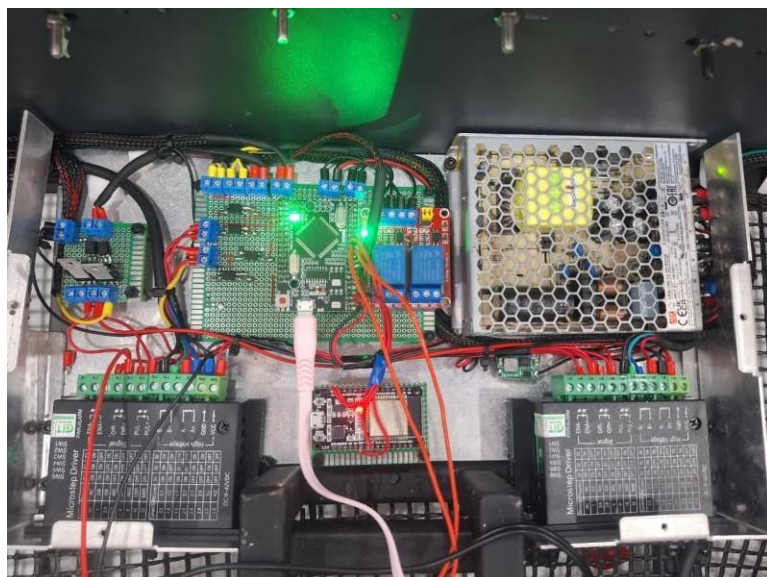


2) Connection and Signal Isolation Testing

After the circuit is installed, the next step is to test the connectivity between devices to ensure that the signal's " " relationship works correctly. Testing begins by ensuring that the PC817 optocoupler is capable of isolating the Arduino Mega and the TB6600 driver, so that the logic signal from the Mega remains clean and is not affected by reverse current surges from the motor side.

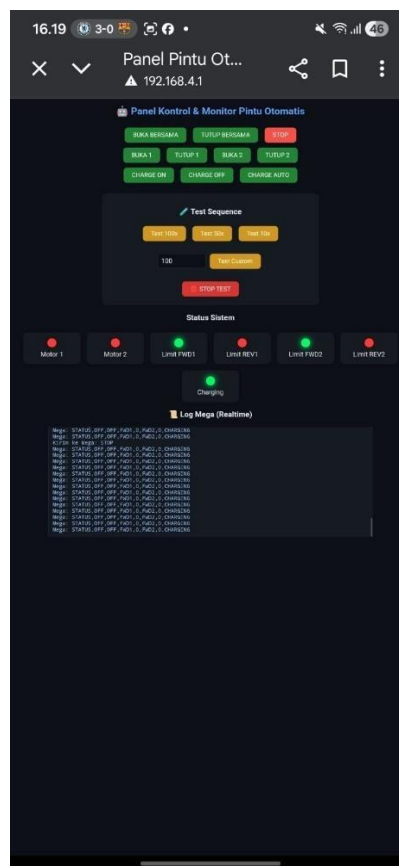


The inspection continues on the limit switch to ensure that changes in its physical condition (pressed or not) result in logical changes that match the program design. On the data communication side, UART testing was performed between the ESP32 and Mega to ensure that commands sent via Serial2 on the ESP32 were received intact by the Arduino Mega. This test also included synchronization of the 115200 baud rate used on both devices.



3) ESP32 and Arduino Mega Software Development

The next step is to develop the software that forms the core of the operating system. On the ESP32 side, a Web Server-based application was created using the ESPAsyncWebServer library, which allows the ESP32 to run as a standalone Access Point with the SSID "ESP32_DOCK_CTRL". The web system was designed using HTML, CSS, and JavaScript to provide an interactive control panel that can display the door status and Arduino Mega logs in real-time. Each button on the web page sends an HTTP GET command such as `/cmd?do=BUKA_BERSAMA` or `/cmd?do=CHARGE_ON`, and the command is forwarded by ESP32 to Arduino Mega via Serial2.



On the Arduino Mega side, the program was developed using the `millis()`-based non-blocking principle so that the stepper motor, limit switch reading, and relay control can run in parallel without delay. The Arduino Mega is also used to send complete status such as motor status,

limit switch condition, and charging status back to the ESP32 in text format so that it can be displayed directly on the web page.

4) Automatic Door Function Testing

After programming was completed, comprehensive testing was conducted on the performance of the double automatic door. Testing began by sending commands from the ESP32 web server such as OPEN_TOGETHER, OPEN_1, CLOSE_2, and STOP to ensure that each motor moved according to the expected response. During testing, it was checked whether the non-blocking function allowed the left and right motors to move synchronously or alternately according to the control logic.

Automatic Door Stops When Open	Door Stops Automatically When Closed
	

The limit switch is tested to ensure that the motor stops automatically when it reaches the open or closed limit, thereby preventing mechanical damage to the door structure. The indicator system on the web is also monitored to ensure that the status LED changes according to the actual condition of the door, such as Motor1 ON, Limit FWD1 active, or Limit REV2 detected. This entire process is validated through real-time logs sent by Arduino Mega and displayed on the web page via the logBox div.

5) Docking Charging System Testing

The final stage is testing the docking charging function controlled by Arduino Mega based on the door status. In automatic mode, the relay is only activated if both doors are completely closed and both motors are stopped. This ensures that the drone charging process is safe, does not interfere with door movement, and does not cause a short circuit. In addition to automatic mode, manual mode testing was also conducted, which can be activated via the CHARGE_ON and CHARGE_OFF commands from the ESP32 web server. All relay activation statuses are also returned to the ESP32 and displayed on the LED indicator marked "Charging". From the test results, the system was proven to be able to regulate the charging process safely and responsively according to the designed logic.

III.6 Results Analysis

The testing was conducted to ensure that the dual automatic door control system and charging docking mechanism based on Arduino Mega and ESP32 could function stably, responsively, and in accordance with the designed logic. The testing covered several important components, namely: stepper motor performance, limit switch accuracy, ESP32–Arduino Mega communication, charging relay function, and status monitoring via a web server. The following are the results of the analysis of each testing stage:

1) Analysis of Stepper Motor Performance in the Door System

The test results show that both HUIYINN 57DNG508Y-21D20 stepper motors can move properly when receiving PUL and DIR signals from Arduino Mega through the TB6600 driver. Door movement in the OPEN_TOGETHER and CLOSE_TOGETHER modes can run synchronously without any significant differences in the opening/closing time between doors. When the doors were tested in individual modes such as OPEN_1, OPEN_2, CLOSE_1, and CLOSE_2, both motors responded to commands accurately without excessive vibration. This proves that the millis()-based non-blocking control method allows for parallel and stable stepper motor movement control.

2) Limit Switch Testing Analysis

The limit switches used as the open limit (FWD) and close limit (REV) worked well when tested in various scenarios. When the door reached the open limit, the FWD signal became active and the system automatically stopped the motor movement. The same thing happens at the close limit, where the REV signal detects the end position and the motor stops without the need for manual intervention. No mechanical noise was found to cause the limit switch to trigger inaccurately, as the system is equipped with a debounce function on the Arduino Mega. These results show that the limit switch is capable of maintaining the mechanical safety of the door and preventing damage due to overtravel.

3) ESP32 – Arduino Mega Serial Communication Analysis

Testing the UART communication between the ESP32 and Arduino Mega shows that the data exchange process runs stably and consistently. All commands sent through the web interface—such as OPEN_TOGETHER, CLOSE_TOGETHER, STOP, CHARGE_ON, and CHARGE_AUTO—are successfully forwarded by the ESP32 and received well by the Arduino Mega without any interference or data loss. Conversely, status data sent from the Arduino Mega to the ESP32 was also received continuously and displayed on the monitoring panel via the *logBox* element on the web page. The use of a baud rate of 115200 proved to be stable enough to handle two-way data traffic, both for sending commands and status updates.

However, there are certain conditions where the system response appears to slow down slightly, especially when the *logBuffer* size starts to increase or when the browser has to re-render a large number of logs every 500 ms. This process causes the ESP32 and browser to need a little extra time to process and display the data, causing users to experience a slight lag in the log display and LED indicator updates. This lag is not caused by interference in UART communication, but is more influenced by the processing load on the web interface side and the fairly fast refresh mechanism. Even so, data accuracy is maintained and communication between the ESP32 and the Arduino Mega does not experience errors or *data corruption*, so overall the system is still reliable and works as needed.

4) Performance Analysis of the ESP32 Web Server and Real-Time Monitoring

The web server developed using the ESP32 module and ESPAsyncWebServer library showed responsive and stable performance during the testing process. All command buttons on the web interface were able to be executed quickly and immediately forwarded to the Arduino Mega without significant delays, allowing users to run door control and charging system functions in real-time. The monitoring mechanism also works well; the virtual LED indicators for the Motor, Limit FWD/REV, and Charging status are updated based on the latest data sent by the Arduino Mega via serial communication.

The data update process, which is performed every 500 ms, is generally fast and accurate, but there are certain conditions where the log display in the browser experiences slight lag. This is not caused by the performance of the ESP32, but rather by the rendering process of the increasingly long log, which requires additional time for the browser to display the text and perform repeated auto-scrolling. However, the lag that appears is only visual and does not affect the communication path or the execution of system commands.

Overall, the ESP32 web server remains capable of performing control and monitoring functions simultaneously with high efficiency. The system has proven to be stable, responsive, and reliable as an operational solution for double automatic doors and WiFi-based docking charging mechanisms, with only a slight decrease in performance on the display side when the log volume is very large.

5) Analysis of the Automatic Charging Docking System

Testing of the charging mechanism shows that the relay only activates when both doors are fully closed. The CHARGE_AUTO mode has been proven to function correctly, where the relay will not activate if one of the doors has not reached the REV state. In addition, the charging system can also be controlled manually using the CHARGE_ON and CHARGE_OFF commands, which facilitate the debugging and testing process. All tests show that the use of a 2-channel relay complies with system safety standards to avoid the risk of short circuits in drone batteries.

6) Overall System Test Results

Based on the entire series of tests that have been carried out, the Arduino Mega and ESP32-based dual automatic door control system and charging docking mechanism performed as designed. Both stepper motors can move stably and responsively, while the limit switch is able to accurately detect the open and close limits, thereby preventing mechanical collisions. Two-way communication between the ESP32 and Arduino Mega also works well; all commands sent via the web server are successfully received by the Arduino, and the system status can be consistently sent back to be displayed on the monitoring panel.

The docking charging relay function works according to the designed automatic logic, which is only active when the door is in a safe condition. The web server interface also appears reliable, where the status of the motor, limit switch, and charging can be updated in real-time. Although there is a slight lag in the log display when the buffer size increases, it does not affect the system control performance or serial communication.

Overall, the developed system has met the design objectives and can be considered suitable for use as a prototype door control system at an automatic drone docking station. This system is capable of integrating motor control, limit sensors, automatic charging, and a web interface into a single stable and easy-to-operate platform.

III.6.1 Testing Table for the Dual Automatic Door Control System

A. Motor & Door Command Function Testing

No	Test Command	Initial Condition	Motor Response	Limit Switch Response	Status on Web Server	Result
1	OPEN_1	Door 1 is closed	Motor 1 is moving forward	FWD1 active → motor stops	Motor1 LED ON → OFF, FWD1	Successful

					LED ON	
2	CLOSE_1	Door 1 open	Motor 1 moves backward	REV1 active → motor stops	Motor1 LED ON → OFF, REV1 LED ON	Success
3	OPEN_2	Door 2 closed	Motor 2 moves forward	FWD2 active → motor stops	Motor2 LED ON → OFF, FWD2 LED ON	Successful
4	CLOSE_2	Door 2 open	Motor 2 moves backward	REV2 active → motor stops	Motor2 LED ON → OFF, REV2 LED ON	Success
5	STOP	Motor is running	Both motors stop instantly	—	All motor LEDs OFF	Success

B. Synchronous Mode Testing (Open/Close Together)

No	Test Command	System Logic	Motor Response	Door Delay	Web Status	Results
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6	OPEN_T OGETHE R	Motor1 running → pause 1.3 seconds → Motor2 running	Both motors move in sequence	1300 ms accord ing to the progra m	Motor status ON alternately → FWD ON	Successf ul
7	CLOSE_T OGETHE R	Motor2 runs → pause 1.3 seconds → Motor1 runs	Both motors move in sequence	1300 ms accord ing to the progra m	Motor status ON alternately → REV ON	Successf ul

C. Limit Switch Test (Mechanical Safety)

No	Test Item	Scenario	System Observation	Result
8	FWD1 Limit	Motor 1 approaching open limit	Motor stops < 120 ms after switch is activated	Success
9	Limit REV1	Motor 1 is approaching the closing limit	Motor stops < 120 ms	Success
10	Limit FWD2	Motor 2 is approaching the open limit	Motor stops < 120 ms	Success
11	REV2 Limit	Motor 2 is approaching the closing limit	Motor stops < 120 ms	Success

D. Docking Charging System Testing

No	Test Mode	Door Condition	Relay Status	Web Status	Result
12	CHARGE_AUTO	Both REV doors active	Relay ON automatically	Charging LED ON	Successful
13	CHARGE_AUTO	One door is not yet REV	Relay remains OFF	Charging LED OFF	Successful
14	CHARGE_ON	Manual mode	Relay ON	Charging LED ON	Successful
15	CHARGE_OFF	Manual mode	Relay OFF	Charging LED OFF	Successful

E. ESP32 ↔ Arduino Mega Communication Test

No	Test	Observation	Result
16	Command transmission from Web → Mega	All commands /cmd?do= are received without delay	Successful
17	Status transmission from Mega → ESP32	The STATUS,... format is received every 500–1000 ms	Successful
18	Real-time log	No buffer overflow, stable scrolling	Successful
19	Web indicator LED	LED color changes according to motor status & limit	Successful
20	Web UI response time to button clicks	Occasionally, there is a 0.5–1 second delay before the command is sent when the logBuffer is very full	Successful with note (slight lag)

21	Data reception time for STATUS from Arduino	Normal updates every 500–1000 ms, but under high load conditions can reach 1500–2000 ms	Successful with a small delay
22	Render LED indicators on the Web UI	The indicator changes according to the status, but with a slight delay when the network is weak/ /RSSI is low	Successful
23	Real-time log performance in the browser	When the log is too long (>3500 characters), scrolling and text parsing slow down slightly	Successful with notes
24	UART communication stability	No data packet loss, but UART flush becomes slightly slow when ESP32 sends logs simultaneously	Stable

BAB IV

CONCLUSION AND RECOMMENDATIONS

IV.1. Conclusion

Based on the series of design, implementation, and testing activities carried out during the Field Work Practice, the following conclusions were obtained:

1. **Successfully designed and built a dual automatic door control system** using Arduino Mega 2560 as the main control center that regulates the movement of two stepper motors, reads the limit switch conditions, and controls the relay for the charging docking mechanism. All electronic circuits such as the TB6600 driver, PC817 optocoupler, limit switch, and relay were successfully integrated functionally.
2. **The communication integration between ESP32 and Arduino Mega via UART ran stably**, allowing ESP32 to function as a web server that received commands from users and displayed the system status in real-time. All commands, such as *OPEN_TOGETHER*, *CLOSE_TOGETHER*, *STOP*, *CHARGE_ON*, and others, were successfully sent to Arduino Mega and executed properly.
3. **The ESP32-based web server is capable of displaying real-time monitoring** such as motor status, limit switch conditions, and charging status without significant interference. Although there is a slight lag in the log display when the data buffer becomes large, this condition does not affect the overall mechanical performance or serial communication.

4. **The system as a whole has functioned according to the design**, where the stepper motor moves stably, the limit switch detects movement limits accurately, the charging relay works according to automatic logic, and web-based control can run responsively. Thus, this system is suitable for use as a prototype for an automatic door mechanism and charging docking for drone stations.

Based on these results, it can be concluded that this project has successfully achieved its design objectives, namely creating a safe, responsive, and integrated automatic door control system via a WiFi-based web interface.

IV.2. Recommendations

Based on the design and testing results of the dual automatic door control system and docking charging mechanism based on Arduino Mega and ESP32, there are several suggestions that can be applied to improve the performance and readiness of the system for use in larger-scale applications:

1. Integration of a GSM Module for Control and Monitoring via a Mobile Application.

As a further development, the system can be upgraded by adding a GSM module such as SIM800L, SIM7600, or A7670 so that the device can connect directly to the cellular network without relying on local WiFi. With a GSM connection, the automatic door system can be controlled and monitored via a mobile application anytime and anywhere. The use of GSM also makes the system more flexible for placement in remote locations or areas that do not have stable internet access.

2. Buffer Management and Log Optimization.

The monitoring system can be improved by limiting the log size on the ESP32 and moving the main log storage to an external server (e.g., Raspberry Pi), so that the WebUI remains responsive and status updates run more smoothly.

3. Addition of Error Handling and System Protection Features.

Development can be carried out by adding detection of abnormal conditions such as motor stall, unreadable limit switches, or UART

communication interference. The implementation of a watchdog timer can also increase the system's resistance to hangs.

4. Use of Encoder Sensors for Stepper Motors.

For higher precision and to ensure no step loss occurs, it is recommended to add an encoder sensor so that the door position can be monitored absolutely and is safer for long-term use.

5. Integration of IoT Systems and Remote Monitoring.

Using a server such as Raspberry Pi, the system can be expanded with MQTT, WebSocket, or an IoT dashboard so that users can monitor the door status and charging process remotely via a local network or the internet.

6. Hardware Design Optimization and Power Isolation.

It is recommended to use a more stable regulator module, additional optoisolation, and a specially designed PCB to minimize noise from the motor driver and improve the safety of logic components.

7. Enhanced Web Access Security.

HTTPS, login authentication, basic firewalls, and IP/MAC-based access control can be added if the web server is run on a more powerful device such as Raspberry Pi. This will be more secure than the limited ESP32 AP mode.

With the implementation of these recommendations, the automatic door system and docking charging system will have better performance, enhanced security, and be ready to be developed into a true industrial prototype.

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ATTACHMENT

1. **Certificate of Completion of Internship (If applicable)**
2. **Work Practice Activity Form**

- 3. Internship Permission Letter from the Department**
- 4. Internship Acceptance Letter from the Company**
- 5. Work Experience Activity Documentation**